MODEL B / END-TO-END REPORT

Supplier-share movement

Data, classes, probability calibration and supplier-risk interpretation

2010-2026 | 1,188 supplier-month rows | H3 / H6

Four probabilities describe collapse, fall, stable or rise in supplier share. Saved forecasts beat both Brier baselines, but roughly half the most-likely classes are incorrect.

ASSESSMENT / READ BEFORE USE

Research / demo. Some disruption and post-disruption calibration gates fail.

Inside: data sources and preprocessing; features and targets; chronological splits; trained and selected models; validation, disruption and post-disruption comparisons; prediction outputs; graphs; reproducibility and limitations.

This is a formatted PDF edition of the packaged README. The figures are embedded from saved model visualizations. No models, datasets or forecasts were changed. Source documentation: 13 October 2026. PDF generated: 15 October 2026.

PDF bookmarks provide section navigation. Relative file links refer to the ready-to-go package; keep this PDF at its root to retain those references.

Updated 13 October 2026 from the saved calibrated-V2 code, data, selections and evaluation files. No fitting, thresholds or selected models were changed.

Summary And Readiness

Model B predicts whether each main supplier's share of UK titanium imports by weight will **collapse, fall, remain stable or rise**, at **H3 and H6**. It returns four probabilities adding to one and the most probable class. It is supervised multiclass classification, not an exact import-volume forecast.

This package beats both required Brier baselines in all six saved validation/test groups. However, calibration fails for **H3 disruption and H3/H6 post-disruption**. Exact class accuracy is approximately **46-56%**, not 80%. It is useful for a clearly caveated research/demo comparison, but not a fully calibrated production risk signal.

H12 labels remain in the upstream dataset for provenance; **this calibrated package does not train or serve H12**.

Data And Sources

Item	Description
Main input	titanium_model_b_country_month_training_2010_2026.csv
Frequency / coverage	Monthly, January 2010-June 2026
Shape	1,188 rows, 75 columns; 198 months x six supplier groups
Supplier groups	USA, Japan, France, Russian Federation, China and Other countries
Product scope	Combined HS 810820 and 810890; HS 810830 waste/scrap excluded
Trade	UN Comtrade: UK import weight, value and prepared supplier shares
Currency	Bank of England Database: exchange-rate conversion and FX movement
Broad price proxy	ONS price statistics: ERUC/PPI
Not included	Required aluminium/nickel benchmark changes remain absent under the earlier source restriction; no fabricated benchmark series were added

Country quantities were **reconstructed from prepared HS supplier shares multiplied by HS net weights**, then summed across the two HS codes. Country share is country quantity divided by total quantity. Monthly country shares sum to one. Reconciliation to the enriched upstream file is not independent validation against original raw partner downloads.

Fixed thresholds, source audit and upstream reconstruction input preserve provenance. The upstream file is used for reconciliation, not passed as an additional predictor matrix. Full raw download-to-panel regeneration is not bundled.

Preparation And Target Labels

The original CSV is retained unchanged. The local loader parses dates, computes extra features separately within each supplier's historical sequence, checks unique country-month keys, verifies shares and aligns targets. Missing feature values are handled by fitted preprocessing; future missing labels are excluded from training.

For horizon h:

share_change[t,h] = country_share[t+h] - country_share[t]

The actual class is derived from this future observed share change; the predicted class is the largest model probability. It is an absolute share difference, not a relative percentage change in country volume. For example, a share changing from 20% to 17% is -3 percentage points, not -3% of its initial share.

Horizon	Fitted classification label	Supporting actual columns
H3	target_class_h3	target_country_share_h3, target_share_change_h3, target_month_h3
H6	target_class_h6	target_country_share_h6, target_share_change_h6, target_month_h6

The future labels are supervised ground truth, never estimator inputs. H3/H6 models are direct-horizon classifiers, not recursive monthly predictions.

Fixed Class Boundaries

Boundaries were derived from pooled pre-test H3 share changes from available 2010-June 2021 history, then frozen. The requested 2005-2009 history is missing. The threshold file records 810 usable changes; these are not 810 distinct months.

Class	Boundary in share-fraction units	Approximate percentage-point interpretation
Collapse	change < -0.1446075178264113	A fall larger than 14.4608 percentage points
Fall	-0.1446075178264113 <= change < -0.022944922950439307	Fall between about 14.46 and 2.29 percentage points
Stable	-0.022944922950439307 <= change <= 0.02088710731068383	Change from about -2.2945 to +2.0887 percentage points
Rise	change > 0.02088710731068383	Increase larger than 2.0887 percentage points

A country with less than 14.46% current share cannot lose over 14.46 percentage points. The selected feasibility rule therefore gives some collapse probabilities exactly zero. This follows share bounds; it is not a missing prediction. It also makes the definition hard to use as a generic collapse detector for small-share countries.

Features And How Models Learn

H3 uses **66 expanded inputs**; H6 uses **35 core inputs**. The full exact lists are in results/model_manifest.json. The feature engine adds derived fields in memory; they need not all exist as columns in the original CSV.

Core inputs	Meaning
country_share; log1p_country_quantity_kg	Current share and transformed supply scale
country_share_change_1m, _3m, _6m, _12m	Historical absolute share movement
country_quantity_log1p_change_1m, _3m, _6m, _12m	Historical transformed country-volume movement
total_import_log_change_1m, _3m, _6m, _12m	Total UK titanium volume movements
titanium_price_log_change_1m, _3m, _6m, _12m	Historical titanium import-price changes, not a live call to Model A
fx_log_change_1m, _3m, _6m, _12m	Exchange-rate movements
eruc_log_change_1m, _3m, _6m, _12m	Broad non-ferrous import-price movements
supplier_hhi_weight; top_supplier_share_weight; partner_count_total	Market concentration and diversification
month_sin; month_cos	Calendar seasonality
eu_collection_change_flag; weight_reconstructed_any_flag; world_weight_estimated_any_flag	Collection and data-quality controls
country_key	Supplier identity; one-hot encoded, not treated as a numeric country ranking

Suffix abbreviations share the displayed prefix. These 35 fields are the H6 input set.

Additional expanded inputs	Derivation / role
country_share_lag_1m, _3m, _6m, _12m	Share at those earlier months, not an average
year_trend	Elapsed months since January 2010
country_share_rank; country_share_relative_to_top; country_share_gap_to_top; is_top_supplier	Current share rank, ratio/gap to the top share and dominance flag
share_velocity_1m, _3m, _6m	Historical share changes normalized per month
share_acceleration_1m_vs_3m, _3m_vs_6m, _6m_vs_12m	Differences between short- and longer-term velocities
quantity_velocity_1m, _3m; quantity_acceleration_1m_vs_3m	Country-volume velocity and acceleration

Additional expanded inputs	Derivation / role
country_share_rolling_mean_3m, _6m, _12m	Recent mean share
country_share_rolling_std_3m, _6m, _12m	Population standard deviation of recent shares
country_share_vs_rolling_mean_3m, _6m, _12m	Current share minus its recent mean
country_share_rolling_range_3m, _6m, _12m	Recent maximum minus minimum share
last_observed_3m_class	Class of the already-observed historical 3-month change, not the future target

Rolling features include the current observed month and use available historical windows with at least two observations. Derived features are deterministic calculations, not random synthetic values. Timing flags are explicit assumptions/eligibility controls rather than model targets.

Forest trees learn conditional splits in historical features; leaf class frequencies produce probabilities. H3 averages Random Forest and Extra Trees probabilities. Temperature scaling changes their sharpness, and the share-bound rule masks impossible classes before renormalizing. Historical-frequency blending was tested but selected weight is zero. Preprocessing fits numeric imputation and categorical encoding using each fitting fold's training rows.

Chronological Split And Calibration

Stage	Origins / outcomes	Interpretation
Initial history	2010 onward; initial validation fits use eligible pre-2018 history	Lag completeness and assumed outcome availability restrict fitting rows
Validation blocks	2018, 2019, January 2020-June 2021	Outcomes must target June 2021 or earlier
Actual validation origins	H3: January 2018-March 2021; H6: January 2018-December 2020	234 and 216 supplier-month predictions respectively
Disruption test	Origins July 2021-June 2023	24 origins x six suppliers = 144 predictions per horizon
Post-disruption	Origins July 2023 onward	H3 ends March 2026 (198 rows); H6 ends December 2025 (180 rows)
Latest inference	June 2026 origin	Six suppliers x two horizons = 12 forecasts

H3 disruption targets run October 2021-September 2023; H6 targets January 2022-December 2023. Post targets end June 2026. These are origin-based tests, not a requirement that every target month lies inside the named period.

Fits expand monthly during testing and use only prior targets assumed available at that origin. Settings are selected using validation, not test rankings, and then frozen. Earlier experiments already inspected the historical test periods, so these are reused evaluations rather than a new untouched holdout.

FX availability is assumed at month-end +7 days; PPI at month-end +31 days; trade timing is inherited. Historical publication dates and revision vintages are unverified. Class thresholds and probability calibration settings are selected using pre-test history, but validation calibration results are not a separate untouched probability-calibration test.

Algorithms Compared And Selected

The experiment compared seven classifiers: core Random Forest, core Extra Trees, core Logistic Regression, supplier-history Extra Trees, expanded Random Forest, expanded Extra Trees and core HistGradientBoosting. It also compared original-core and expanded forest ensembles.

For each candidate it tested temperatures 0.75/1.0/1.25/1.5/2.0, historical-frequency blend weights 0/0.15/0.3/0.5 and share feasibility on/off. Selection prefers configurations beating both Brier baselines and having maximum calibration gap <=0.10, then minimizes validation Brier, with ECE/log-loss tie breakers. All configurations.

Horizon	Selected model	Inputs	Probability settings	Saved artifact
H3	Expanded Random Forest + Extra Trees ensemble	66	Temperature 1.25, frequency blend 0, feasibility enabled	model_b_h3_calibrated.joblib
H6	Core Extra Trees classifier	35	Temperature 0.75, frequency blend 0, feasibility enabled	model_b_h6_calibrated.joblib

Selection is not a claim that these algorithms are universally best. Each bundle includes preprocessing, estimators, fixed thresholds, configuration and historical frequencies.

Metrics And Acceptance

Metric	Interpretation
Accuracy	Fraction where the largest-probability class equals actual_class
Balanced accuracy	Mean recall across classes; reduces dominance of common classes
Macro F1	Mean class F1; balances precision/recall without weighting by class frequency
Multiclass Brier	Mean sum of four squared probability errors; lower is better, range 0-2 under this implementation, not divided by four
Log loss	Penalizes low probability assigned to the actual class
Expected calibration error (ECE)	Count-weighted average probability-versus-frequency gap
Maximum calibration gap	Worst occupied bin gap; a sparse bin can fail despite small average ECE
Brier skill versus frequency	1 - model Brier / historical-frequency Brier; above zero is better

Baselines are **last-observed-change class** (one-hot persistence probabilities) and **supplier historical class-frequency probabilities** fitted to eligible history. They are not random predictions.

Reliability flattens the four class probabilities and binary actual indicators into ten probability bins. Empty bins have no observed frequency; occupied sparse bins remain in the test. The saved acceptance rule requires beating both Brier baselines and maximum occupied-bin gap <=0.10. It does **not** require 80% accuracy.

Main Performance Table

Period	H	Rows	Accuracy	Balanced accuracy	Macro F1	Brier	Frequency baseline	Persistence baseline	Max gap	Result
Validation	H3	234	48.72%	40.32%	0.4063	0.6323	0.7100	1.5299	0.0901	Pass
Validation	H6	216	45.83%	46.09%	0.4548	0.6300	0.7005	1.4537	0.0553	Pass
Disruption	H3	144	52.78%	51.09%	0.5322	0.6064	0.6539	1.4306	0.2636	Fail calibration
Disruption	H6	144	47.22%	53.26%	0.4793	0.6218	0.6900	1.4028	0.0826	Pass
Post	H3	198	55.56%	48.95%	0.5054	0.5666	0.6528	1.4545	0.1124	Fail calibration
Post	H6	180	54.44%	62.38%	0.5362	0.5827	0.6766	1.4556	0.5294	Fail calibration

Source: evaluation_metrics.csv.

Period	H	ECE	Log loss
Validation	H3	0.0199	1.0832
Validation	H6	0.0265	1.0892
Disruption	H3	0.0448	1.0274
Disruption	H6	0.0151	1.0727
Post	H3	0.0387	0.9663
Post	H6	0.0318	1.0058

Comparison With Earlier Models

Period / H	Original Brier	Earlier accuracy experiment	Current calibrated V2	Interpretation
Validation H3	0.6428	0.6377	0.6323	Improved Brier
Validation H6	0.6316	0.6457	0.6300	Slight improvement
Disruption H3	0.6025	0.6103	0.6064	Worse than original, better than accuracy experiment
Disruption H6	0.6237	0.6394	0.6218	Slight improvement
Post H3	0.5835	0.5816	0.5666	About 2.9% lower Brier than original, but calibration still fails
Post H6	0.5851	0.6090	0.5827	Slight improvement, calibration still fails

Source: previous_vs_new_metrics.csv. Better probability error is not necessarily better accuracy: post H3 accuracy is about 0.51 percentage points lower than the original model. H6 accuracy is unchanged from the original in these saved periods.

Practical interpretation: these probabilities contain more information than the baselines, but about half the most-likely classes are incorrect. This is not an 80%-accurate supplier forecast. A calibration failure means a stated probability and the observed frequency differ too much in at least one bin, not that every prediction is wrong. Post H6's worst bin contains only five class-probability observations; it still counts as failure. Post H3's failing bin has 64 observations. Reliability bins.

Ground Truth And Forecast Outputs

all_ground_truth_predictions.csv contains validation and both tests. Compare **actual_class** with **predicted_class**; **correct** is their equality. actual_future_share and actual_share_change explain the derived actual label. Confidence is the maximum model probability, **not verified case-level accuracy** or an independent probability that the selected class is correct.

Latest forecasts and response JSON use June 2026 as the origin. H3 targets September 2026; H6 targets December 2026. The computer date does not update this observation cutoff.

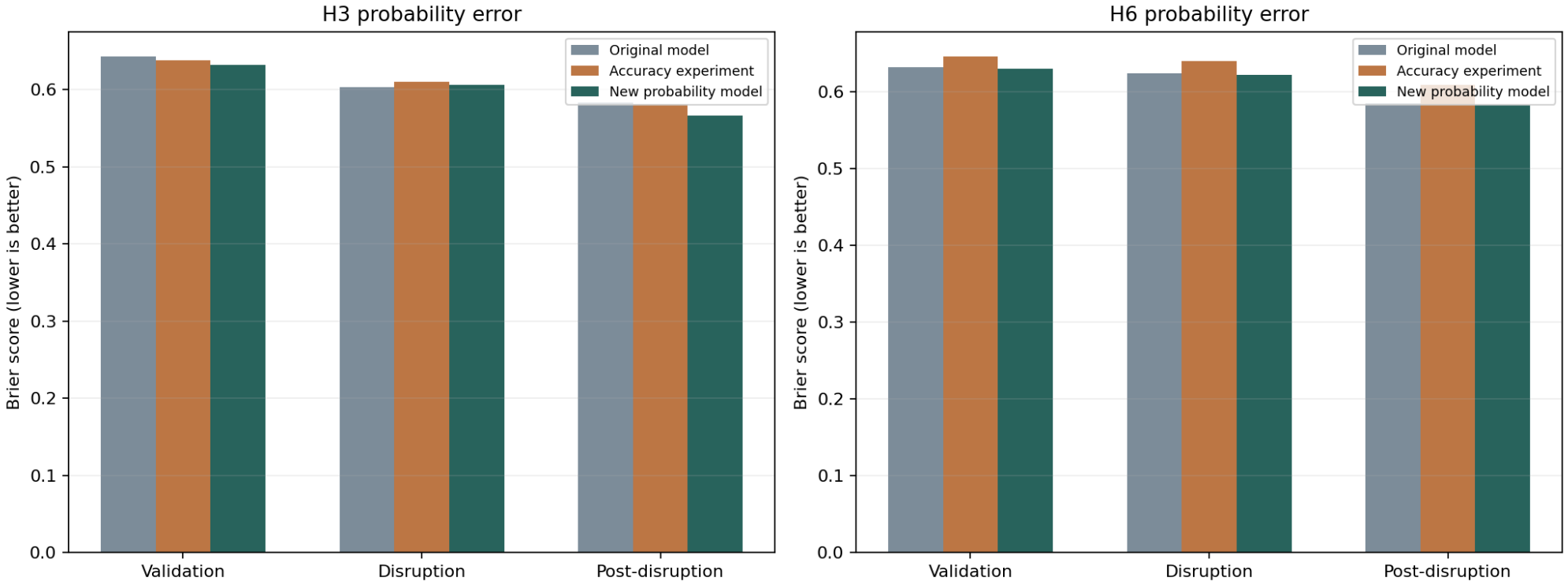
Examples; probabilities are shown as percentages rounded independently:

Supplier	H	Collapse	Fall	Stable	Rise	Most likely
Japan	H3	1.52%	17.23%	10.16%	71.09%	Rise
Russian Federation	H3	0.00%	12.96%	64.46%	22.58%	Stable
USA	H6	54.81%	19.38%	11.82%	13.99%	Collapse

The full-precision four probabilities sum to one; displayed rounded percentages can differ slightly from 100%. The response includes country, current share, horizon, origin/target month, algorithm, class and confidence. It does not predict an exact future country share or volume.

Figure 1. Brier comparison with previous approaches and baselines

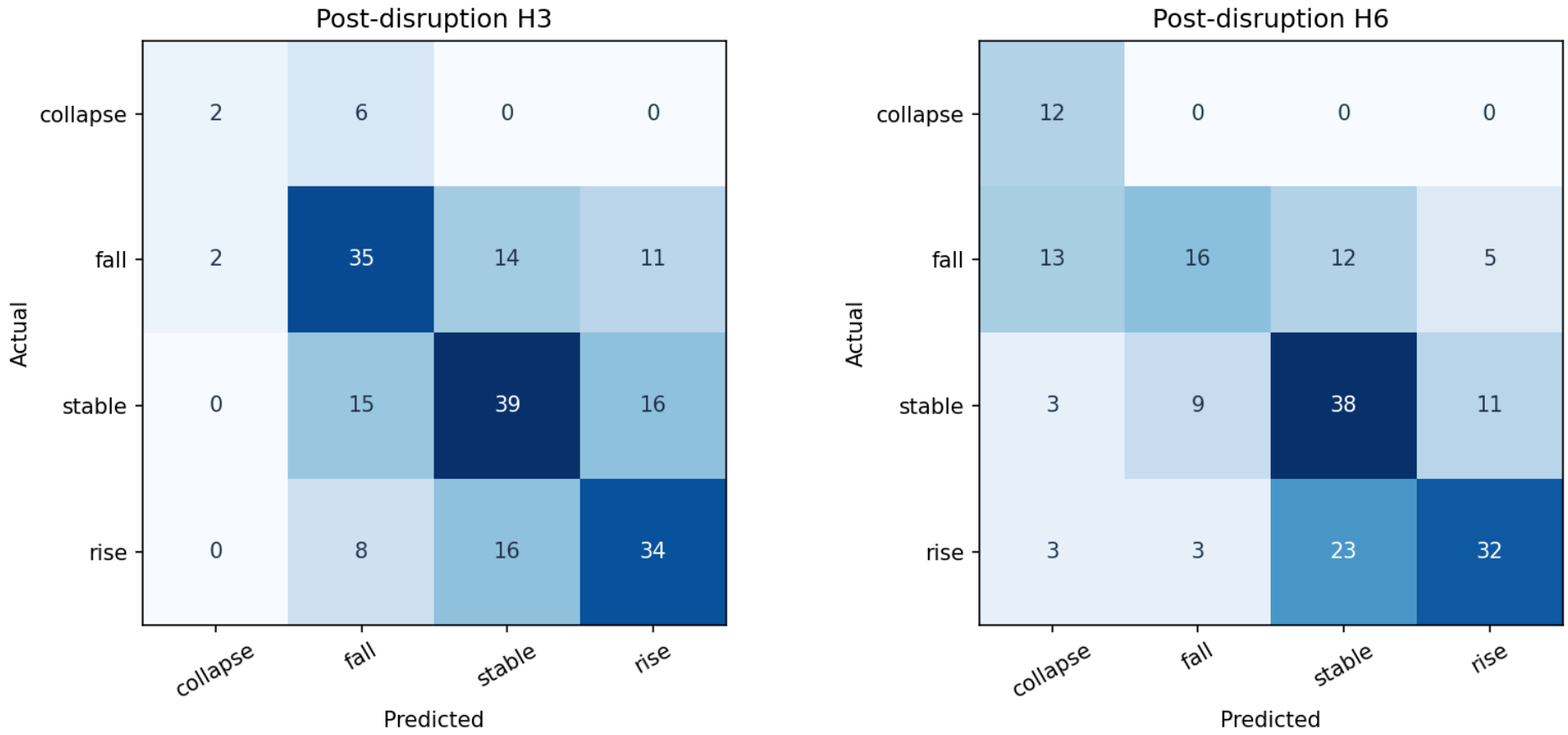
Saved visualization / Graphs And Detailed Diagnostics



Source: visualizations/brier_comparison.png

Figure 2. Actual versus predicted confusion matrices

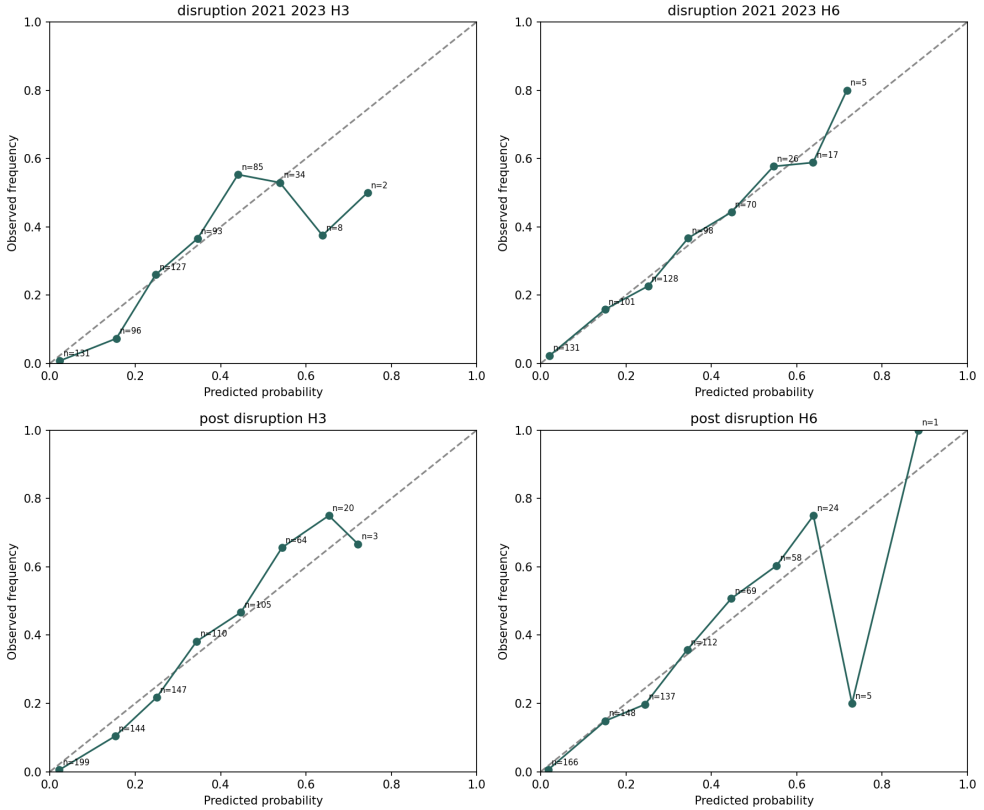
Saved visualization / Graphs And Detailed Diagnostics



Source: visualizations/confusion_matrices.png

Figure 3. Probability reliability

Saved visualization / Graphs And Detailed Diagnostics



Source: visualizations/probability_reliability.png

Russia risk check | Interactive dashboard | Supplier-level metrics | Class recalls.

GitHub renders images; download/open the HTML locally for interactive filters. No learned feature-importance chart is claimed for this selected ensemble merely because a previous experiment has one.

Package Map And Run Commands

Location	Contents
data/	Original training panel, fixed thresholds, feature/selection manifests and reconciliation/comparison inputs
code/	Training/inference, local feature engine and verification scripts
models/	Selected H3/H6 calibrated bundles
results/	Actual/predicted records, metrics, reliability, selection, source audit and latest outputs
visualizations/	PNG charts and standalone HTML dashboard

Run from this calibrated V2 folder:

```
python -m pip install -r requirements.txt
python code/verify_local_data.py
python code/train_probability_model_b.py --predict-only
```

Verification checks the source hash, fixed thresholds and **48 latest probability values**. Predict-only regenerates the latest CSV/JSON, not historical evaluation. The entry point uses local data/ and code/model_b_feature_engine.py, not required parent-folder inputs. The helper feature-engine file is a dependency, not the selected training entry point.

Running train_probability_model_b.py without flags performs training and writes results; preserve the current evidence or use a new experiment folder. The original model source directory was not changed during packaging. Load only trusted joblib files.

Outstanding Requirements And Next Work

Model B still has missing 2005-2009 history, UN Comtrade rather than requested HMRC inputs, absent required aluminium/nickel changes, and hardcoded supplier membership requiring raw-partner verification against the 2018-2020 >=5% selection rule. It reconciles prepared weights, not raw imports. No Russia H3/H6 disruption observation has a collapse target under the frozen boundaries, so the intended Russia-collapse demonstration is unresolved.

Improve calibration with an untouched chronological calibration/holdout study, examine sparse-bin and class/supplier reliability, verify source vintages and partner aggregation, and seek additional genuine predictors rather than manufacturing missing history. Retain the fixed thresholds for fair comparisons. Do not merge bins or select test winners merely to obtain a pass.

B's Brier/classification accuracy cannot be directly compared with A/C regression errors. See the root source/reuse review before public data sharing. The package is technically verified for saved-model inference, not production-approved; clean-environment installation remains untested.